

An Unbounded Additive Gap in a Minimum-Zero Approach to Five-Cycle Double Covers

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Scope warning. This paper does *not* prove or disprove the Five-Cycle Double Cover Conjecture. It disproves an unrestricted auxiliary assertion that was proposed as one possible route to that conjecture. Every graph constructed here has an explicit five-cycle double cover.

Abstract

For a cubic graph, a useful reformulation of the five-cycle double cover problem asks for a matching that is simultaneously the exact zero set of an $\mathbb{F}_2^2$ -flow and the intersection of two binary cycles. A natural proof strategy is to minimize the number of zero edges first and then try to extend a minimum exact-zero matching to such a certificate. We give a certified 130-vertex counterexample to this strategy: its minimum exact-zero matching size is 5, whereas every matching/flow/two-cycle certificate has size at least 6, and one of size 6 exists. We then prove an elementary factorization theorem for two-edge sums. Iterating the sum gives connected simple bridgeless cubic graphs G_n with

$$r_f(G_n) = r_M(G_n) = 5 \cdot 2^n, \quad \eta(G_n) = 6 \cdot 2^n.$$

Thus the additive extension gap is unbounded. The infinite-family argument is human-checkable; the finite base statement is backed by explicit positive witnesses and LRAT certificates checked by two independent proof checkers.

1 Introduction and exact scope

A binary cycle of a finite graph is an edge set having even incidence at every vertex. A standard five-cycle double cover (five-CDC) is an ordered list of five binary cycles such that every edge belongs to exactly two members of the list. Empty members are allowed, so this convention also expresses “at most five”. The Five-Cycle Double Cover Conjecture asserts that every finite bridgeless graph has such a cover. The conjecture is due independently to Celmins and Preissmann; a current exposition of the resolved unbounded-size cycle-double-cover problem explicitly records the five-member strengthening as still open [6].

This note studies only a proposed proof route for cubic graphs. The route starts with an $\mathbb{F}_2^2$ -flow having as few zero edges as possible, requires those zero edges to form a matching, and asks that the matching extend to the intersection of two binary cycles. The unrestricted assertion that some cardinality-minimum exact-zero matching always extends is false. The purpose of this note is to make that negative result, and its amplification under two-edge sums, precise and reproducible.

The result has two parts.

1. A finite, computer-assisted base theorem supplies a 130-vertex graph with extension gap $6 - 5 = 1$.

*Draft prepared with substantial use of AI systems. See the disclosure in Section 9.

2. A self-contained two-edge-sum theorem proves that both minima factor exactly on identical rooted copies. This turns the base graph into an infinite family with unbounded additive gap.

Neither part produces a five-CDC counterexample. Indeed the size-six certificate on the base graph, and its iterated gluing, give explicit five-CDCs.

Publication status. This is a research draft, not a priority claim. The literature comparison in Section 7 is sufficient to delimit the claim made here, but a specialist review and independent human verification of the finite certificate package are still required before submission.

2 Definitions

Throughout, graphs are finite and loopless. The constructed graphs are simple, although the definitions make sense for loopless multigraphs. For an edge set D , write ∂D for the set of vertices incident with an odd number of edges of D . Thus D is a binary cycle exactly when $\partial D = \emptyset$.

Definition 2.1. An $\mathbb{F}_2^2$ -flow on a graph G is a map $\phi : E(G) \rightarrow \mathbb{F}_2^2$ satisfying

$$\bigoplus_{e \ni v} \phi(e) = 0$$

at every vertex v . Its exact zero set is $Z(\phi) = \{e : \phi(e) = 0\}$.

Definition 2.2. For a cubic graph G , define

$$\begin{aligned} r_f(G) &= \min_{\phi} |Z(\phi)|, \\ r_M(G) &= \min_{\substack{\phi \\ Z(\phi) \text{ a matching}}} |Z(\phi)|. \end{aligned}$$

If the second feasible set is empty, set $r_M(G) = +\infty$.

Definition 2.3. A matching/four-flow certificate on G is a quadruple (M, A, B, ϕ) such that

1. M is a matching;
2. A and B are binary cycles;
3. $A \cap B = M = Z(\phi)$; and
4. ϕ is an $\mathbb{F}_2^2$ -flow.

Let

$$\eta(G) = \min\{|M| : (M, A, B, \phi) \text{ is a matching/four-flow certificate}\},$$

with value $+\infty$ if there is no certificate.

Plainly $r_f(G) \leq r_M(G) \leq \eta(G)$ whenever the quantities on the right are finite.

Proposition 2.4 (standard certificate equivalence [4, Corollary 0.6]). *A finite loopless cubic multigraph has a standard five-CDC if and only if it has a matching/four-flow certificate.*

For completeness, the direction used below can be checked directly. Given (M, A, B, ϕ) , put $D_0 = A \triangle B$. On $G - M$, represent the three nonzero values of $\mathbb{F}_2^2$ by

$$s(1) = 1100, \quad s(2) = 1010, \quad s(3) = 0110,$$

and put $k = 1111$. If $\ell(y)$ denotes coordinate zero of $s(y)$, define

$$h(e) = 1_{D_0}(e) + \ell(\phi(e)), \quad d(e) = s(\phi(e)) + h(e)k.$$

Both summands defining h are binary cycles, so d is a four-coordinate flow. Each $d(e)$ has weight two, and its coordinate zero is $1_{D_0}(e)$. If D_0, D_1, D_2, D_3 are the coordinate supports, then

$$A, B, D_1, D_2, D_3$$

double-cover every edge. Conversely, pair-label the edges of a five-CDC; at each cubic vertex the three labels form a triangle in K_5 . Taking the intersection of two fixed coordinates gives the required matching, and quotienting the remaining four even coordinates by $\langle 1111 \rangle$ gives the required $\mathbb{F}_2^2$ -flow.

3 The certified 130-vertex base graph

3.1 Explicit construction

Let Q be the 60-vertex cubic graph whose graph6 encoding is

```
{??????00?B?D?EGGSAK?H_?HG?Ao@A_?????????????C?G?C???M???B_???
[????????????????????0?C??C????B_????M?????[_????????????????????
C??_???0?????B_?????B_?????@q????????????????????????????C???G???
C????????M????????B_????????[_????????????????????????????0????
G???C????????B_????????M?????????]
```

where the displayed line breaks are to be removed. Mark the five edges

$$(22, 26), (30, 34), (38, 42), (46, 50), (54, 58).$$

Take two disjoint copies of Q . Subdivide the five marked edges in each copy, respecting the displayed order, and join corresponding new subdivision vertices. Call the resulting graph G_0 . It has 130 vertices and 195 edges. The five joining edges form a matching.

The ancillary package

`search/minimum-zero-exchange-countermodel-130v-20260727/`

contains the full edge list, graph6 encoding, positive witnesses, two CNF instances, two LRAT proofs, a generator, and an independent verifier.

Theorem 3.1 (certified base theorem). *The graph G_0 is simple, connected, bridgeless, and cubic, and*

$$r_f(G_0) = r_M(G_0) = 5, \quad \eta(G_0) = 6.$$

Moreover G_0 has an explicit standard five-CDC.

Certificate proof. The retained flow has exact zero set equal to the five joining edges. Direct incidence checks verify flow conservation and verify that this zero set is a matching. Thus $r_M(G_0) \leq 5$.

The first CNF asks for an arbitrary $\mathbb{F}_2^2$ -flow with at most four zero edges. Its LRAT proof of unsatisfiability is accepted by both `lrat-check` and CakeML `cake_lpr`. Hence $r_f(G_0) \geq 5$. Since $r_f \leq r_M$, the two inequalities prove

$$r_f(G_0) = r_M(G_0) = 5.$$

The second CNF is the complete matching/four-flow encoding with $|M| \leq 5$. Its LRAT proof is accepted by the same two checkers, so $\eta(G_0) \geq 6$. A retained size-six witness directly verifies

$$M = A \cap B = Z(\phi),$$

the matching condition, the two binary-cycle conditions, and flow conservation. Therefore $\eta(G_0) = 6$. The package also stores the five pair labels obtained by the reverse construction in Proposition 2.4; direct parity checking verifies that the five coordinate supports double-cover every edge.

Finally, the verifier checks simplicity, cubicity, connectedness, and bridgelessness directly from the reconstructed edge list. $\square$

Remark 3.2. The computer-assisted part of Theorem 3.1 is finite and certificate-producing. The positive assertions are checked directly. The two negative assertions rely on LRAT proofs, not on trusting the SAT solver that found them. Appendix 8 gives replay instructions and fixed hashes.

4 Exact factorization across a two-edge sum

Let G_1, G_2 be disjoint connected bridgeless cubic graphs with distinguished edges

$$e_1 = u_1v_1, \quad e_2 = u_2v_2.$$

Delete e_1, e_2 and add

$$f = u_1u_2, \quad g = v_1v_2.$$

Denote the resulting graph by $G_1 \#_2 G_2$. The crossed pairing has the same properties.

Lemma 4.1 (graph structure). *If the factors are connected, bridgeless, and cubic, then $G_1 \#_2 G_2$ is connected, bridgeless, and cubic. If the factors are simple, the sum is simple.*

Proof. Cubicity, simplicity, and connectedness are immediate. Each new edge lies on the cycle consisting of f, g and a u_i -to- v_i path in each $G_i - e_i$; such paths exist because the root edges are not bridges. For an old edge h , choose a cycle of its factor containing h . If that cycle avoids the deleted root edge, it survives. Otherwise, replace the root edge on the cycle by the path through the other factor and f, g . Hence every edge of the sum lies on a cycle, so the sum is bridgeless. $\square$

Lemma 4.2 (cut parity). *Let D be a binary cycle in a graph and let $\{f, g\}$ be an edge cut. Then D contains either both of f, g or neither.*

Proof. Sum the even-degree equations of D over one shore of the cut. Internal incidences cancel, leaving $1_D(f) + 1_D(g) = 0 \pmod{2}$. $\square$

Lemma 4.3 (flow restriction and zero count). *Let $H = G_1 \#_2 G_2$, and let ϕ be an $\mathbb{F}_2^2$ -flow on H . Then $\phi(f) = \phi(g) = a$ for some $a \in \mathbb{F}_2^2$. Restoring e_i with value a gives a flow ϕ_i on G_i , and*

$$|Z(\phi)| = |Z(\phi_1)| + |Z(\phi_2)|.$$

If $Z(\phi)$ is a matching, then each $Z(\phi_i)$ is a matching.

Proof. Sum the flow equations over $V(G_1)$. Every internal edge appears twice and cancels, leaving $\phi(f) + \phi(g) = 0$; in characteristic two this says $\phi(f) = \phi(g) = a$. Assigning a to each restored root edge makes the endpoint equations in each factor agree with the endpoint equations in H .

If $a \neq 0$, neither cut edge nor either restored root is zero. If $a = 0$, both cut edges and both restored roots are zero. In either case, the number of zeros in H is the sum of the numbers in the two restored factors.

For the matching assertion, only the case $a = 0$ needs comment. Since the two zero cut edges meet the four root endpoints, no other zero edge of ϕ meets any of those endpoints. Replacing the two cut edges by one root edge in each factor therefore preserves the matching condition. $\square$

Lemma 4.4 (certificate restriction). *Every matching/four-flow certificate on $H = G_1 \#_2 G_2$ restricts to a matching/four-flow certificate on each factor, and its matching size is the sum of the two restricted matching sizes.*

Proof. Let (M, A, B, ϕ) be a certificate. By Lemma 4.2, each of A, B uses both cut edges or neither. Restore e_i in each factor with the common membership bit for A , the common membership bit for B , and the common flow value supplied by Lemma 4.3. Binary-cycle parity and flow conservation are restored at every root endpoint. On a root edge the identities

$$1_M = 1_A 1_B, \quad 1_M = 1_{\{\phi=0\}}$$

hold because they held with the same states on f and g . Lemma 4.3 preserves the matching condition and proves the additive count. $\square$

Theorem 4.5 (identical-rooted factorization). *Let G be a connected bridgeless cubic graph with a distinguished edge, and let $H = G \#_2 G$ be formed from two identically rooted copies. Then*

$$r_f(H) = 2r_f(G).$$

If $r_M(G) < \infty$, then

$$r_M(H) = 2r_M(G).$$

If $\eta(G) < \infty$, then

$$\eta(H) = 2\eta(G).$$

Proof. Lemma 4.3, applied to an arbitrary flow on H , gives

$$r_f(H) \geq 2r_f(G).$$

Take a minimum flow on G , put identical copies on the two factors, delete the two root edges, and give both new edges the common old root value. The endpoint equations hold and the zero-count identity gives the reverse inequality.

If the zero set is required to be a matching, the same restriction and gluing arguments preserve that condition by Lemma 4.3. This proves the statement for r_M .

Lemma 4.4 gives $\eta(H) \geq 2\eta(G)$. Conversely, glue two identical copies of a minimum certificate. The two cycle-membership bits and the two flow bits on the roots agree by construction, so replacing the roots with f, g restores all parity equations and all certificate identities. The matching count doubles. $\square$

Remark 4.6. For two nonisomorphic factors, the three restriction arguments still give the corresponding additive lower bounds. Equality can fail if minimum objects in the two factors have incompatible states on their chosen root edges. Identical rooted copies avoid this compatibility issue.

5 The infinite family

Set G_0 equal to the graph of Theorem 3.1. Having defined a rooted G_n , form G_{n+1} from two identically rooted copies by the two-edge sum, and choose any surviving edge as the next distinguished root.

Corollary 5.1 (unbounded additive extension gap). *For every $n \geq 0$, the graph G_n is finite, simple, connected, bridgeless, and cubic, with*

$$r_f(G_n) = r_M(G_n) = 5 \cdot 2^n, \quad \eta(G_n) = 6 \cdot 2^n.$$

In particular,

$$\eta(G_n) - r_M(G_n) = 2^n$$

is unbounded. The graph G_n has $130 \cdot 2^n$ vertices and $195 \cdot 2^n$ edges.

Proof. The structural assertions follow inductively from Lemma 4.1. The three formulas follow by applying Theorem 4.5 at every step, starting from Theorem 3.1. A two-edge sum deletes two edges and adds two, so both the vertex and edge counts double. $\square$

Corollary 5.2. *Every G_n has a standard five-CDC.*

Proof. The finite value of $\eta(G_n)$ supplies a matching/four-flow certificate. Apply Proposition 2.4. $\square$

6 What the construction does not show

The family is a counterexample family only to the unrestricted minimum-zero extension strategy. It is not a counterexample family to the Five-Cycle Double Cover Conjecture.

For $n \geq 1$, the top-level composition exposes a cyclic two-edge cut. The 130-vertex base graph itself has girth 4 and cyclic edge-connectivity 3. Consequently this construction does not address a version of the auxiliary assertion restricted to cyclically four-edge-connected, high-girth graphs arising in standard minimum-counterexample reductions.

The result establishes an unbounded *additive* gap. The ratio $\eta(G_n)/r_M(G_n) = 6/5$ is constant, so no claim of an unbounded multiplicative gap is made.

The base theorem is computer-assisted. The infinite factorization proof does not depend on a solver, but its unconditional numerical starting values depend on the correctness of the explicit finite witnesses, CNF encodings, LRAT certificates, and proof-checker implementations.

7 Related work and novelty boundary

The matching/four-flow characterization is not new. Hoffmann-Ostenhof proved that a cubic graph has a five-CDC if and only if it has a matching M , a nowhere-zero 4-flow on $G - M$, and two cycles whose exact intersection is M [4, Corollary 0.6]. Zhang gave a broader characterization of five-cycle double covers through covering subgraphs with nowhere-zero 4-flows [7], and Liu, Hao, Luo, and Zhang developed a 4-flow and Catlin-reduction approach for five-CDCs [5]. Proposition 2.4 is therefore included as the exact certificate language used by the computation, not as a new existence characterization.

Flow resistance was formalized as an edge-uncolourability measure by Fiol, Mazzuoccolo, and Steffen [3]. Allie, Máčajová, and Škoviera constructed snarks with resistance n and flow resistance $2n$ [1], and later showed that the ratio between those two established parameters can be arbitrarily large [2]. Those results concern r_f versus ordinary resistance. The parameter η here instead measures the least exact-zero matching that also extends to the two-cycle certificate required for a five-CDC.

In the sources checked for this draft, we did not find the parameter η , the assertion $r_M = \eta$, or the exact two-edge-sum factorization in Theorem 4.5. The defensible contribution is therefore narrow: a certificate-backed counterexample to that auxiliary assertion and a human-checkable composition theorem making the additive gap unbounded. This is an apparent novelty statement, not a claim of priority. A domain expert should still check older flow, snark superposition, and cycle-double-cover literature before submission.

8 Reproduction and certificate boundary

From the project root, run

```
cd search/minimum-zero-exchange-countermodel-130v-20260727
shasum -a 256 -c SHA256SUMS
python3 verify.py
```

The verifier reconstructs the graph independently from the 60-vertex core and marked edges, checks graph structure and every positive witness, regenerates both CNFs independently, compares them clause-for-clause with the retained instances, and invokes both `lrat-check` and CakeML `cake_lpr` on both retained LRAT proofs.

The critical retained hashes are:

Artifact	SHA-256
<code>graph.json</code>	<code>a0cea6013ed45534441d072b8579792a643a6ad3b67f0948ec8a70eb0c5156b2</code>
<code>witness.json</code>	<code>3dd958be4f8a6dbd88c2fbdca04e9904562468193db5e0c74f51f885adcfde79</code>
<code>flow-at-most-four-zero.cnf</code>	<code>60c5c6a46aebf646d6d55c633052df56aa459aaf28a144b343a58bda0bf08397</code>
<code>flow-at-most-four-zero.lrat</code>	<code>462544fce1c1d70797445136eaa9319aba159e42ae622ce9d229c662a909d626</code>
<code>extension-at-most-five.cnf</code>	<code>df5f7e54866626c39c2f9c5a91b51a1ac4024174413707c1c700a4f319b6a6c8</code>
<code>extension-at-most-five.lrat</code>	<code>e4a87af680d1a33d72f7e6825ab2f31f6a600eb87d200461776df51c5b5dd687</code>

The first CNF has two flow-bit variables and one zero-indicator per edge. It enforces both vertex-parity equations, the equivalence

$$z_e \longleftrightarrow (\neg p_e \wedge \neg q_e),$$

and $\sum_e z_e \leq 4$.

The second CNF has variables

$$m_e, a_e, b_e, p_e, q_e$$

per edge. It enforces

$$m_e \longleftrightarrow (a_e \wedge b_e), \quad m_e \longleftrightarrow (\neg p_e \wedge \neg q_e),$$

matching clauses $\neg m_e \vee \neg m_{e'}$ for incident pairs, four vertex-parity equations for a, b, p, q , and $\sum_e m_e \leq 5$. Parity is expanded by its complete falsifying truth table; the cardinality constraints use a sequential counter. Thus satisfiability of the second instance is exactly existence of a matching/four-flow certificate of size at most five.

The additional script

```
preprint/minimum-zero-exchange-gap/audit_two_sum.py
```

independently checks the positive base witnesses and, for every one of the 195 possible root edges, constructs the two-copy sum and checks the glued size-ten minimum-flow witness, the glued size-twelve certificate, and the glued five-CDC.

9 AI-use disclosure

This research and this manuscript were produced with substantial assistance from OpenAI Codex agents operating under human direction. AI systems participated in conjecture formulation, computational search, construction discovery, SAT and certificate encoding, proof exploration, independent-code drafting, audit suggestions, and substantial portions of the prose and L^AT_EX drafting. The two-edge-sum proof was proposed and checked in this AI-assisted process; it should not be represented as a solely human derivation.

The use of AI does not replace mathematical verification. The intended trust boundary is explicit:

1. The factorization theorem and infinite-family corollary are elementary arguments written in full in this paper and should be checked line by line by human readers.
2. Positive finite claims reduce to public graph and witness data and direct parity checks.
3. Negative finite claims reduce to fixed CNF files and fixed LRAT proofs. They are replayed by two proof checkers, one of which is the CakeML verified checker used in the project package.
4. The manuscript has not yet received independent specialist peer review. Human verification, literature review, authorship review, and venue-specific AI disclosure are required before submission.

Any public version should preserve this disclosure and should list the exact AI systems and versions if the submission venue requests them.

References

- [1] I. Allie, E. Máčajová, and M. Škoviera, Snarks with resistance n and flow resistance $2n$, *Electron. J. Combin.* 29 (2022), no. 1, Paper P1.44, <https://doi.org/10.37236/10633>.
- [2] I. Allie, E. Máčajová, and M. Škoviera, Flow resistance to resistance ratios in cubic graphs, *Discrete Appl. Math.* 342 (2024), 362–367, <https://doi.org/10.1016/j.dam.2023.09.026>.
- [3] M. A. Fiol, G. Mazzuoccolo, and E. Steffen, Measures of edge-uncolourability of cubic graphs, *Electron. J. Combin.* 25 (2018), no. 4, Paper P4.54.
- [4] A. Hoffmann-Ostenhof, A note on 5-cycle double covers, *Graphs Combin.* 29 (2013), 977–979, <https://doi.org/10.1007/s00373-012-1169-8>.
- [5] S. Liu, R.-X. Hao, R. Luo, and C.-Q. Zhang, 5-cycle double covers, 4-flows, and Catlin reduction, *SIAM J. Discrete Math.* 37 (2023), 253–267, <https://doi.org/10.1137/22M1472425>.
- [6] S.-I. Oum, A proof of the cycle double cover conjecture by OpenAI: an exposition, arXiv:2607.16356v2 (2026), <https://arxiv.org/abs/2607.16356>.
- [7] C.-Q. Zhang, Nowhere-zero 4-flows and cycle double covers, *Discrete Math.* 154 (1996), 245–253, [https://doi.org/10.1016/0012-365X\(95\)00047-Z](https://doi.org/10.1016/0012-365X(95)00047-Z).